# COS deep-dive — the priority file

Fang invented COS. If she asks one hard question, it is probably here. Goal: I can do everything on the checklist **closed-book, on paper, out loud**.

## Mastery checklist (tick when you can do it cold)

- ☐ Derive the COS pricing formula for a European option from the Fourier-cosine expansion of the density.
- ☐ State where the characteristic function enters and why COS only needs the cf, not the density.
- ☐ Explain why a **cosine** expansion (even extension, real coefficients) rather than a full complex Fourier series.
- ☐ Write the payoff coefficients $V_k$ for a call/put and explain the closed form.
- ☐ Choose the truncation range [a, b] from cumulants: $a, b = c_1 \pm L\cdot\sqrt{c_2 + \sqrt{c_4}}$. Explain each term.
- ☐ Explain the convergence rate: exponential for smooth densities, algebraic otherwise — and what kills it.
- ☐ **My Assignment 2 bug:** why a small c2 (short maturity) makes [a, b] too narrow to cover the spot grid, and exactly how I fixed it.
- ☐ COS for Heston: plug in the Heston cf; state the Feller condition.
- ☐ COS for Barrier options (Lecture 10) — how the method adapts.
- ☐ COS for American options (Lecture 9) — the approach used.
- ☐ COS vs Carr–Madan FFT: when and why COS wins.

## High-probability questions (have Claude Code grill me on these)

1. Derive the COS European call price. Where does the cf enter?

   **A.** Start from the discounted-expectation price in the log-state $y$ (e.g. $y = \ln(S_T/K)$):

   $$P(x_0,T) = e^{-rT} \int_{\mathbb{R}} V(y,T)\, f(y|x_0)\, dy.$$

   I do not know $f$, but I know its characteristic function $\varphi_{X_T}(t;x_0) = E[e^{i t X_T}|x_0] = \int e^{ity} f(y)\, dy$, i.e. $\varphi$ is the Fourier transform of $f$.

   **Step 1 — truncate.** Pick $[a,b]$ (cumulant rule, Q2) containing essentially all the mass and restrict the integral to $[a,b]$.

   **Step 2 — cosine-expand the density.** On $[a,b]$ write the half-range cosine series (prime halves $k=0$):

   $$f(y) \approx {\sum_{k\geq 0}}' A_k \cos\left(k\pi \frac{y-a}{b-a}\right), \quad A_k = \frac{2}{b-a} \int_a^b f(y) \cos\left(k\pi\frac{y-a}{b-a}\right) dy.$$

   **Step 3 — $A_k$ from $\varphi$ (the engine).** Write $\cos = \mathrm{Re}\{\exp\}$, pull out the phase $e^{-ik\pi a/(b-a)}$, and extend the integral to $\mathbb{R}$ (the *only* approximation — range truncation):

   $$A_k \approx F_k = \frac{2}{b-a} \mathrm{Re}\left\{ \varphi\left(\frac{k\pi}{b-a}; x_0\right) \cdot \exp\left(-ik\pi \frac{a}{b-a}\right) \right\}.$$

   The remaining $\int_{\mathbb{R}} f(y)\, e^{i(k\pi/(b-a))y}\, dy = \varphi(k\pi/(b-a))$ exactly. No integral survives.

   **Step 4 — payoff coefficients.** Define $V_k = \frac{2}{b-a} \int_a^b V(y,T) \cos\left(k\pi\frac{y-a}{b-a}\right) dy$. For the call $V = K(e^y-1)^+$ lives on $y\geq 0$, so

   $$V_k^{call} = \frac{2}{b-a} K \left[ \chi_k(0,b) - \psi_k(0,b) \right].$$

   **Step 5 — assemble.** Insert the expansion, swap $\Sigma/\int$ (uniform convergence of the cosine series for smooth $f$), and use $V_k$:

   $$P \approx e^{-rT} \frac{b-a}{2} {\sum_{k=0}^{N-1}}' A_k V_k.$$

   Since $\frac{b-a}{2} \cdot A_k = \mathrm{Re}\{\varphi(k\pi/(b-a);x_0)\, e^{-ik\pi a/(b-a)}\}$, this is

$$P \approx e^{-rT} \sum'_{k=0}^{N-1} \mathrm{Re}\{ \varphi(k\pi/(b-a); x_0) \exp(-ik\pi\, a/(b-a)) \}\, V_k.$$

**Where the cf enters:** in exactly one place — as the cosine coefficients of the density, $\mathrm{Re}\{\varphi \cdot e^{-ik\pi a/(b-a)}\}$. The spot dependence sits inside $\varphi(\cdot;x_0)$; for BS/Lévy $\varphi(u;x_0)=\varphi_{lvl}(u)\, e^{iux_0}$, so one set of $\{V_k\}$ prices the whole spot/strike grid (only the cheap $e^{iux_0}$ changes). Cost $O(N)$, no integration, no FFT.

2. How do you pick [a, b]? What breaks at short maturity, and how did you fix it in your assignment?

   **A. Cumulant rule (Fang–Oosterlee):**

   $$[a,b] = [\, c_1 - L\sqrt{c_2+\sqrt{c_4}},\ \ c_1 + L\sqrt{c_2+\sqrt{c_4}}\, ], \quad L \in [8,12] \text{ (typ. 10)},$$

   where $c_1$ = mean (centres the box), $c_2$ = variance ($\sqrt{c_2}$ is the natural width scale, so $L\sqrt{c_2}$ is an "L-sigma" box), $\sqrt{c_4}$ (fourth cumulant) fattens the box for heavy tails / kurtosis (zero for Gaussian), and $L$ is the safety multiplier — larger $L$ lowers the truncation floor but needs more terms.

   **BS cumulants** of $\ln(S_T/K)$:

   $$c_1 = \ln(S_0/K) + (r - \tfrac{1}{2}\sigma^2)T, \quad c_2 = \sigma^2 T, \quad c_3 = c_4 = 0 \quad \Rightarrow \quad [a,b] = c_1 \pm L\sigma\sqrt{T}.$$

   **What breaks as $T\to 0$:** $c_2 = \sigma^2 T \to 0$, so the half-width $L\sqrt{c_2+\sqrt{c_4}} \to 0$ and the box collapses toward the single point $c_1$. With $c_4=0$ (Gaussian) there is no $\sqrt{c_4}$ floor to keep it open. Two failure modes: (i) the box gets too narrow to cover where $V\cdot f$ has mass; (ii) when pricing a whole grid of spots/strikes, a range centred by $c_1$ for one $(S_0,K)$ is centred away from other grid points, so the cosine reconstruction is evaluated where it is meaningless — large, erratic errors at short maturity.

   **Diagnostic signature** (this is the tell): the error *grows* as $T\to 0$ and does *not* improve when $N$ increases. That distinguishes a *range* error (a fixed accuracy floor) from a *series* error (curable by more terms).

   **Assignment-2 fix:** the truncation range was too narrow at short maturities because $c_2$ was tiny. I (i) raised $L$ at short $T$, (ii) imposed a minimum half-width floor so $b-a$ cannot collapse (equivalently kept a $\sqrt{c_4}$-type term), and (iii) made sure the range was computed to cover the entire spot/strike grid actually being priced, not a single point. After that the error stopped growing as $T\to 0$ and reverted to exponential decay in $N$.

3. Why cosine series instead of a complex Fourier series?

   **A.** Three reasons:

   - **(i) Real coefficients / real arithmetic.** The even extension of $f$ on $[a,b]$ kills every sine term, so the surviving $A_k$ are real ($F_k$ is literally a $\mathrm{Re}\{\cdot\}$). A full complex series carries twice the coefficients and complex bookkeeping for no gain.
   - **(ii) Direct sampling from φ.** The cosine coefficient *is* the real part of $\varphi$ evaluated at the grid frequency $k\pi/(b-a)$: $A_k = (2/(b-a))\, \mathrm{Re}\{\varphi(k\pi/(b-a))\, e^{-ik\pi a/(b-a)}\}$. The cosine basis is exactly the one whose coefficients are read off the cf in one line — no inversion integral.
   - **(iii) Payoff side closes in elementary functions.** $V_k$ are integrals of (polynomial/exponential)×cos, which integrate to the closed forms $\chi_k$, $\psi_k$. With a complex-exponential basis the call coefficients are not as clean. The whole method pairs the cosine coefficients of $f$ (from $\varphi$) against those of $V$ (closed form).

4. What sets the convergence rate? Give a case where it is only algebraic.

   **A.** Three error sources: (1) **range truncation** $[a,b]$ — *fixed* once chosen, it sets the accuracy *floor* at which convergence saturates (exponentially small in the half-width for exponentially-decaying tails); (2) **series truncation** at $N$ — governed by the decay of the cosine coefficients $A_k$; (3) **cf/model error** — feeds straight through.

   The rate in $N$ is set by the **smoothness of $f$**:

- $f$ analytic in a strip / $f \in C^\infty$ with integrable derivatives ⇒ $A_k$ decay **exponentially** ⇒ **exponential** convergence in $N$ (machine precision in dozens–low hundreds of terms).
- $f$ with only $\beta$ integrable derivatives ⇒ $A_k = O(k^{-(\beta+1)})$ ⇒ only **algebraic** convergence. *Lack of smoothness is what kills the exponential rate.*

**Algebraic case:** a low-regularity density — e.g. Variance-Gamma with small $\nu$ (the density develops a non-smooth peak / cusp), or a short-dated density approaching a Dirac, or near a jump/kink. There $A_k$ decay only polynomially and you need many more terms.

5. Write the Heston characteristic function you used as ground truth. State the Feller condition.

**A.** Heston: $dS = rS\,dt + \sqrt{v}\,S\,dW1$, $dv = \kappa(\bar{v} - v)dt + \eta\sqrt{v}\,dW2$, $d\langle W1,W2\rangle = \rho\,dt$, with $\kappa$ mean-reversion speed, $\bar{v}$ long-run variance, $\eta$ vol-of-vol, $\rho$ correlation, $v0$ initial variance. The characteristic function of the log-price $x_T = \ln(S_T/S0)$ (or $\ln S_T$) is affine-exponential:

$$\varphi(u;T) = \exp\Big( i u (\ln S0 + rT) + (v0/\eta^2) \cdot \big( (1 - e^{-DT})/(1 - G e^{-DT}) \big) \cdot (\kappa - i\rho\eta u - D) + (\kappa \bar{v}/\eta^2) \cdot \big( (\kappa - i\rho\eta u - D)T - 2 \ln( (1 - G e^{-DT})/(1 - G) ) \big) \Big),$$

with

$$D = \sqrt{ (\kappa - i\rho\eta u)^2 + (u^2 + iu)\,\eta^2 },$$
$$G = (\kappa - i\rho\eta u - D) / (\kappa - i\rho\eta u + D).$$

$D$ and $G$ come from solving the Riccati ODE for the variance factor of the affine system. (Equivalent "little-trap" form uses $G$ rather than $g=1/G$ to keep the complex log on the principal branch and avoid branch-cut discontinuities — the cf must be evaluated continuously or COS picks up $\varphi$-error that feeds straight through.)

**Feller condition:** $2\kappa\bar{v} \geq \eta^2$. It guarantees the CIR variance process stays strictly positive (never hits 0), which keeps the density well-behaved and $\varphi$ clean; if violated, $v$ can touch zero and the simulation/quadrature near $v=0$ needs care.

6. How does COS extend to Barrier options? To American options?

**A. Barrier (Lecture 10), discretely monitored on $t_1<\ldots<t_M=T$.** Knock-out survival is a product of indicators $\prod_m 1_{\{x_m<h\}}$ ($h = \ln(H/K)$). By the **Markov property** the joint density factorises into one-step transitions, so the $M$-dimensional integral nests into a **backward recursion**. Define continuation and option value at each date:

$$c(x,t_{m-1}) = e^{-r\Delta t} \int v(y,t_m)\, f(y|x)\, dy,$$
$$v(x,t_{m-1}) = R \text{ (knocked out) for } x \geq h, \quad = c(x,t_{m-1}) \text{ (alive) for } x < h.$$

Each continuation integral is *exactly* the European COS sum:

$$\hat{c}(x,t_{m-1}) = e^{-r\Delta t} {\textstyle\sum'_k} F_k(x)\, V_k(t_m), \quad F_k(x)=\mathrm{Re}\{\varphi(k\pi/(b-a);x)\, e^{-ik\pi a/(b-a)}\}.$$

The new ingredient: the "payoff" $v(y,t_m)$ is produced by the previous step, so its coefficients $V_k$ are **propagated by backward induction**. Because $v$ is piecewise at $h$, $V_k$ splits at the barrier:

$$\hat{V}_k(t_m) = \hat{C}_k(a,h,t_m) + e^{-r(T-t_{m-1})}\, (2R/(b-a))\, \Psi_k(h,b),$$

and $\hat{C}_k$ (inserting the COS continuation) becomes $\sum_j \varphi_{lvl}(j\pi/(b-a))\, V_j\, M_{\{k,j\}}$ with an analytic coupling $M_{\{k,j\}}$. $M_{\{k,j\}}$ splits into a **Hankel** part ($j+k$) and a **Toeplitz** part ($j-k$); Toeplitz/Hankel×vector = convolution = 3 FFTs by circulant embedding ⇒ $O(N \log N)$ per step, total $O((M-1)N \log_2 N)$. So — unlike European COS — the barrier method *does* use the FFT, because it must move the whole coefficient vector at every step. "In" barriers come from in–out parity. Continuous monitoring is the $M\to\infty$ limit, recovered by Richardson extrapolation in $\Delta t$ using the known $O(\sqrt{\Delta t})$ discrete-vs-continuous bias. Heston ⇒ 2-D state (log-stock, variance): 1-D COS in log-stock + numerical integration over variance, or full 2-D COS; Feller $2\kappa\bar{v}\geq\eta^2$.

**B. American (Lecture 9).** Same backward-recursion idea but the decision is *early exercise* rather than a barrier knock-out. On a time grid, at each step the option value is $v(x,t_m) =$

$\max(g(x), c(x,t_m))$ — the larger of immediate exercise payoff $g$ and the COS continuation value $c(x,t_m)=e^{-r\Delta t}\Sigma'_k F_k(x)V_k(t_{m+1})$. The early-exercise boundary $x^*_m$ is the point where $g = c$; below/above it (call vs put) the option is exercised. So each step (i) finds $x^*_m$ (a 1-D root-find on $g - c = 0$), (ii) splits the cosine coefficients of $v$ at $x^*_m$ into a continuation region (cosine coeffs of $c$, computed from the previous $V_k$ via the same analytic coupling integrals) and an exercise region (cosine coeffs of $g$, closed-form $\chi_k/\psi_k$). This is the COS Bermudan pricer; the American price is obtained by Richardson extrapolation in the number of exercise dates. Structurally identical to the barrier recursion, with the moving boundary $x^*_m$ playing the role the fixed barrier $h$ plays there.

7. Compare COS with Carr–Madan. Advantages and limitations.

   **A.** Both need only the characteristic function. Carr–Madan damps the call ($c_T(k)=e^{\alpha k}C_T(k)$, $\alpha>0$, to make it $L^1 \cap L^2$), Fourier-transforms it in closed form in terms of $\varphi$, and recovers the price by an **inverse FFT** — i.e. a trapezoidal-type quadrature of an oscillatory integrand, $O(N \log N)$.

| | COS | Carr–Madan / FFT |
|---|---|---|
| Core op | one $O(N)$ cosine sum, no integration | FFT quadrature, $O(N \log N)$ |
| Free params | none (range from cumulants) | damping $\alpha$ must be tuned |
| Convergence (smooth $f$) | exponential in $N$ | algebraic (quadrature-limited) |
| Terms for ~machine prec. | dozens–hundreds | thousands of nodes |
| Greeks | analytic, same sum | extra transforms / finite differences |

**COS wins** for smooth/piecewise-smooth densities with known $\varphi$ and exponentially-decaying tails — essentially all European pricing under BS, Heston, and standard Lévy/AJD models — because it converges exponentially, needs no free parameter, and gives Greeks for free.

**Limitations / be careful:** genuinely low-regularity densities (slow algebraic $A_k$ decay), very short maturities, or fat tails, where $[a,b]$ must be chosen carefully (Q2) and $N$ raised; and any error in $\varphi$ (e.g. a mis-evaluated Heston cf across the branch cut) feeds straight through. Carr–Madan's FFT also naturally returns a *whole strip of strikes* in one transform, which COS matches only because its $\{V_k\}$ are strike-independent in log-moneyness.

## My answers (concise board versions — re-derive these out loud, don't read)

Compressed versions of the full answers above; expand any of them on demand.

1. **COS call price.** $P = e^{-rT}\int V f \, dy$; I don't know $f$ but know its cf $\varphi$. Truncate to $[a,b]$, cosine-expand $f$, and the coefficients come straight off $\varphi$: $A_k \approx (2/(b-a))\mathrm{Re}\{\varphi(k\pi/(b-a))e^{-ik\pi a/(b-a)}\}$ (extend the coefficient integral to $\mathbb{R}$ — the only approximation). Pair against payoff coeffs $V_k=(2/(b-a))\int_a^b V \cos(...)dy$; the integral of the product is $(b-a)/2\ \Sigma' A_k V_k$, giving $P \approx e^{-rT}\ \Sigma'_{k=0}^{N-1} \mathrm{Re}\{\varphi(k\pi/(b-a);x_0)e^{-ik\pi a/(b-a)}\}\ V_k$. The cf enters only as the cosine coefficients of the density; $x_0$ lives inside $\varphi$, so one $\{V_k\}$ prices the whole grid.

2. **$[a,b]$.** $c_1 \pm L\sqrt{c_2+\sqrt{c_4}}$, $L\approx 10$: $c_1$ centres, $\sqrt{c_2}$ is the width scale, $\sqrt{c_4}$ fattens for fat tails, $L$ is the safety margin. BS: $c_1=\ln(S_0/K)+(r-\tfrac{1}{2}\sigma^2)T$, $c_2=\sigma^2 T$, $c_4=0$, so $[a,b]=c_1\pm L\sigma\sqrt{T}$. As $T\to 0$, $c_2\to 0$ so the box collapses (no $\sqrt{c_4}$ floor in the Gaussian case) → error *grows* as $T\to 0$ and is *not* cured by more $N$ (that signature = range, not series, error). Assignment-2 fix: raise $L$ at short $T$, impose a minimum half-width floor, and size the range to the whole spot/strike grid, not one point.

3. **Why cosine.** Even extension ⇒ real $A_k$, real arithmetic, half the storage; $A_k$ is literally $\mathrm{Re}\{\varphi\}$ at the grid frequency (direct cf sampling); payoff coeffs reduce to elementary $\chi_k$, $\psi_k$. A complex series gives none of these cleanly.

4. **Convergence.** Three errors: range (fixed floor), series (decay of `A_k`), cf/model (passes through). Rate set by smoothness of `f`: analytic ⇒ exponential `A_k` decay ⇒ exponential in `N`; only β integrable derivatives ⇒ `A_k=O(k^{-(β+1)})` ⇒ algebraic. Algebraic case: VG with small ν, or short-dated density near a Dirac/kink.
5. **Heston cf + Feller.** Affine-exponential cf in `u` with `D=√((κ-iρηu)²+(u²+iu)η²)`, `G=(κ-iρηu-D)/(κ-iρηu+D)`, variance term `(v0/η²)((1-e^{-DT})/(1-Ge^{-DT}))(κ-iρηu-D)` plus the `(κv̄/η²)` drift term; use the "little-trap" form for a continuous branch. Feller: `2κv̄ ≥ η²` keeps `v>0` and φ clean.
6. **Barrier / American.** Both = backward recursion via Markov factorisation; each step is European COS `ĉ=e^{-rΔt}Σ'F_k(x)V_k(t_m)` with `V_k` propagated. Barrier: knock out by overwriting `v=R` above `h`, `V_k` splits at `h`, the coupling `M_{k,j}` is Hankel+Toeplitz ⇒ FFT, `O((M-1)N log₂N)`. American: `v=max(g,c)`, split `V_k` at the early-exercise boundary `x*_m` (root of `g-c`); Bermudan + Richardson in #exercise dates. The moving boundary `x*` plays the role of the fixed barrier `h`.
7. **COS vs Carr–Madan.** Both need only φ. CM damps the call (param α) and inverts by FFT quadrature — algebraic, `O(N log N)`, tunable α. COS is a single `O(N)` cosine sum, no integration, no free parameter, exponential convergence, free Greeks. COS wins for smooth densities with decaying tails; be careful for low-regularity / very short `T` / fat tails (set `[a,b]`, raise `N`), and any φ error feeds straight through.

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# Lecture 6 drill set (European COS)

Companion to `notes/Lecture06-notes.pdf`. Each question has a terse answer key in the collapsible note — cover it and answer out loud first. "Derive" means on paper, no notes.

## A. Density recovery

**A1.** Write the half-range cosine expansion of a density f on [a,b], with the prime convention. What does the prime mean and why is A0/2 the mean of f over [a,b]? > Key: f(x)=Σ' A_k cos(kπ(x-a)/(b-a)), A_k=(2/(b-a))∫_a^b f cos(kπ(x-a)/(b-a))dx. Prime halves k=0. A0 = (2/(b-a))∫f = (2/(b-a))·(mass), and A0/2 over the constant basis function is the average height.

**A2.** *(Core derivation.)* Derive A_k ≈ (2/(b-a)) Re{ φ(kπ/(b-a)) exp(-ikπa/(b-a)) } starting from the coefficient integral. Identify the single approximation you make and where. > Key: cos = Re of exp; pull out e^{-ikπa/(b-a)}; the remaining ∫_a^b f e^{i(kπ/(b-a))x}dx is extended to ℝ (THE approximation — range truncation) = φ(kπ/(b-a)). Everything else exact.

**A3.** Standard normal: φ(t)=e^{-t²/2}. Write F_k for a=-b symmetric. What range do the slides suggest and why (10^{-8} clip)? > Key: substitute φ(t)=e^{-t²/2} at t=kπ/(b-a) into F_k, giving F_k=(2/(b-a))Re{e^{-½(kπ/(b-a))²} e^{-ikπa/(b-a)}}. The slides set a=Φ^{-1}(1e-8), b=-a, i.e. clip each tail at probability 1e-8 — symmetric because the standard normal is symmetric about 0. The Gaussian factor e^{-½(kπ/(b-a))²} makes F_k decay super-exponentially in k, which is the canonical exponential-convergence picture.

## B. Why cosine

**B1.** Three reasons COS uses a cosine series, not a complex Fourier series. Be specific about each. > Key: (i) even extension ⇒ real coefficients, real arithmetic, half the storage; (ii) A_k is literally Re{φ at grid frequency} — direct sampling from ch.f.; (iii) payoff coefficients V_k reduce to elementary χ_k, ψ_k (exp/poly × cos integrate in closed form).

## C. Pricing formula

**C1.** *(Core derivation.)* From P = e^{-rT}∫ V f dy, derive P ≈ e^{-rT} Σ'_{k=0}^{N-1} Re{φ(kπ/(b-a);x0) e^{-ikπa/(b-a)}} V_k. State the V_k definition and the sum–integral swap justification. > Key: insert cosine expansion of f, swap Σ/∫ (uniform convergence, smooth f), define V_k=(2/(b-a))∫_a^b V cos(…)dy ⇒ P≈e^{-rT}(b-a)/2 Σ' A_k V_k; sub (b-a)/2·A_k = Re{φ e^{-ikπa/(b-a)}}; truncate at N.

**C2.** Where exactly does the ch.f. enter, and where does the spot x0 dependence live? Why does one set of {V_k} price the whole strike/spot grid? > Key: φ enters only as the cosine coefficients of the density, Re{φ e^{-ikπa/(b-a)}} — nowhere else. The spot x0 sits inside φ(·;x0); for Lévy/BS φ(u;x0)=φ_lvl(u)e^{iux0}, so across a strike/spot grid only the cheap e^{iux0} phase changes while the level cf φ_lvl is reused. The payoff coeffs V_k are strike-independent in the log-moneyness variable, so they are computed once and reused for the whole grid — that is the source of COS's grid efficiency.

**C3.** What is the per-price computational cost of the COS sum? Does it call an FFT? > Key: O(N) per price — a single finite cosine sum of N terms, no numerical integration and no FFT. Each term is one cf evaluation times a precomputed V_k, so the whole price is essentially a dot product. This is the headline contrast with Carr–Madan (O(N log N) FFT quadrature) and with the discrete-barrier COS (which does need the FFT because it propagates the full coefficient vector at every step).

## D. Payoff coefficients V_k

**D1.** *(Derive.)* Define χ_k(c,d)=∫_c^d e^y cos(ω_k(y-a))dy, ω_k=kπ/(b-a). Derive the closed form. > Key: χ_k=Re{e^{-iω_k a}/(1+iω_k) [e^{(1+iω_k)y}]_c^d}; rationalise by (1-iω_k)/(1+ω_k$^2$); χ_k = 1/(1+ω_k$^2$)[cos(ω_k(d-a))e^d - cos(ω_k(c-a))e^c + ω_k sin(ω_k(d-a))e^d - ω_k sin(ω_k(c-a))e^c].

**D2.** Write ψ_k(c,d) including the k=0 case. Why must k=0 be special-cased? > Key: for k≠0, ψ_k(c,d)=(1/ω_k)[sin(ω_k(d-a))-sin(ω_k(c-a))]=(b-a)/(kπ)[sin(ω_k(d-a))-sin(ω_k(c-a))]. For k=0, ω_0=0 so the integrand cos(0)=1 and ψ_0=d-c. The k=0 case must be special-cased because the general 1/ω_k formula is 0/0 there (a removable singularity); in code a naive division by ω_0=0 gives NaN, so the k=0 term — which also carries the ½ from the prime — is set explicitly.

**D3.** In y=ln(S_T/K), give V_k for call and put. Why is the call interval [0,b] and the put interval [a,0]? > Key: call V_k=(2/(b-a))K[χ_k(0,b)-ψ_k(0,b)] — payoff K(e$^{y-1)}$+ lives on y≥0; put V_k=(2/(b-a))K[ψ_k(a,0)-χ_k(a,0)] — payoff K(1-e$^{y)}$+ lives on y≤0.

## E. Truncation range (THE bug)

**E1.** State [a,b]=[c1 - L√(c2+√c4), c1 + L√(c2+√c4)]. Interpret each of c1, c2, √c4, L. > Key: c1 mean (centres), c2 variance (√c2 width scale, L-sigma box), √c4 fattens box for heavy tails/kurtosis (0 for Gaussian), L safety multiplier (8–12, typ. 10).

**E2.** Give the BS cumulants of ln(S_T/K) and the resulting box. Which cumulants vanish? > Key: c1=ln(S0/K)+(r-½σ$^2$)T, c2=σ$^2$T, c4=0 (and c3=0) ⇒ [a,b]=c1 ± Lσ√T.

**E3.** *(High priority.)* As T→0, what happens to [a,b] and why? Give the diagnostic signature distinguishing a range problem from a series problem. How do you fix it? > Key: c2=σ$^2$T→0 ⇒ half-width→0 ⇒ box collapses to ~c1; with c4=0 no floor keeps it open. Signature: error GROWS as T→0 and does NOT improve when N increases (range error is a floor, not curable by more terms). Fix: raise L at short T; keep √c4 / impose a minimum half-width floor; ensure the range covers the whole spot/strike grid actually priced, not one point.

## F. Convergence

**F1.** List the three error sources in the COS price and say which one sets the accuracy floor. > Key: (1) range truncation [a,b] — FIXED once chosen, sets the floor; (2) series truncation at N — set by decay of $A_k$; (3) ch.f./model error feeds straight through. Convergence saturates at the range floor.

**F2.** What property of f gives exponential vs only algebraic convergence in N? Give a case where it is only algebraic. > Key: smoothness/analyticity ⇒ $A_k$ decay exponentially ⇒ exponential in N. Finite smoothness (β integrable derivatives) ⇒ $A_k=O(k^{-(\beta+1)})$ ⇒ algebraic. Algebraic case: density with low regularity / near a kink/jump, e.g. VG with small $\nu$, or short-dated densities approaching a Dirac.

**F3.** COS vs MC convergence rate? Roughly how many terms for machine precision on Heston (cf slide 31)? > Key: MC converges at $O(N^{-1/2})$ in the number of paths, whereas COS converges exponentially in the number of terms for smooth densities. On Heston (slide 31) N~160 terms reach ~1e-10 error, versus thousands of FFT nodes for worse error and longer CPU time. So COS reaches machine precision with N in the dozens to low hundreds — the gap is the whole selling point of the method.

## G. Greeks

**G1.** Why are Delta and Vega “free” in COS? Write the Delta sum. > Key: S0 (via $x=\ln(S0/K)$) and the model parameters enter the COS sum only analytically, through $\varphi(\cdot;x0)=\varphi(\cdot)e^{iux}$; so differentiation passes straight through the finite sum with no re-integration. $\Delta\approx e^{-rT}\Sigma' \, \mathrm{Re}\{\varphi(k\pi/(b-a)) \, e^{ik\pi(x-a)/(b-a)} \cdot ik\pi/(b-a)\} V_k/S0$ (chain rule $\partial x/\partial S0=1/S0$). Vega differentiates φ w.r.t. the variance parameter instead. The same $\{V_k\}$ are reused, so Greeks cost essentially nothing extra — a structural advantage over quadrature/FFT pricers that need finite differences.

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# Lecture 10 drill set (COS for barrier options)

Companion to `notes/Lecture10-notes.pdf`. Same cover-and-answer format.

## H. The pricing problem

**H1.** Define up-and-out vs up-and-in call. Why are there exactly 8 barrier types? > Key: up-and-out call pays $\max(S_T-K,0)$ UNLESS the asset crosses B>S0 during [0,T], in which case it pays 0 (knocked out); up-and-in call pays 0 UNLESS B is crossed, in which case it becomes a vanilla call (knocked in). The 8 types = up/down × in/out × call/put (2×2×2). Each has a closed-form GBM price.

**H2.** State in–out parity and give its two practical uses. > Key: holding knock-in and knock-out with the same (K,B,T) reproduces the vanilla, because exactly one of the two is alive at maturity: $C_{in} + C_{out} = e^{-r(T-t)}E[(S_T-K)^+]$. Use (i): price the easier leg (usually the knock-out) plus the vanilla, then back out the other by subtraction — cheaper and more stable than handling the in-condition directly. Use (ii): it gives a free correctness check on any numerical barrier price.

**H3.** Write the two equivalent formulations (expectation and PDE). Where does the barrier enter the PDE? > Key: $V=e^{-r(T-t)}E_x[(\alpha(e^{X_T}-K))+ 1_{\tau_B>T}]$; or localized Feynman–Kac BS PDE on [a,b] with terminal payoff and ABSORBING boundary V(a,t)=V(b,t)=0. Barrier enters via the homogeneous Dirichlet BC, not the equation.

## I. Reflection principle (derive)

**I1.** *(Core.)* Derive $P(\tau_a \le T) = 2P(W_T \ge a)$ and hence the density of the running maximum Ξ. > Key: split $P(\tau_a\le T)$ by $\{W_T>a\},\{W_T\le a\}$; reflection ⇒ both halves equal ⇒ $=2P(\tau_a\le T,W_T\ge a)=2P(W_T\ge a)$ (since $\{W_T\ge a\}\subseteq\{\tau_a\le T\}$). So $\Xi \sim |W_T|$, $P(\Xi\le a)$=erf-type, $p_\Xi(a)=\sqrt{2/\pi T}\, e^{-a^2/2T}\, 1_{\{a\ge 0\}}$.

**I2.** *(Core.)* Derive the joint density of $(\Xi_0^T, W_T)$. > Key: for $b \le a$, reflect after $\tau_a$: $P(\Xi \ge a, W_T<b)=P(\Xi \ge a, W_T>2a-b)=P(W_T>2a-b)$ (since $2a-b \ge a$). Differentiate $-\partial^2/\partial a \partial b \Rightarrow p(a,b)=\sqrt{2/\pi}(2a-b)/T^{3/2}\, e^{-(2a-b)^2/2T}\, 1_{a \ge \max(b,0)}$.

**I3.** How do you get from standard BM to drifted BM (GBM log-price) results? > Key: Girsanov change of measure. A drifted BM $W_T+\theta t$ under P is a driftless BM under an equivalent measure Q, with Radon–Nikodym derivative $dP/dQ = e^{\theta W_T - \frac{1}{2}\theta^2 T}$. So I compute the reflection-principle result for driftless BM under Q, then reweight by that exponential to recover the drifted (GBM log-price) law. This is exactly what turns the driftless joint density of $(\Xi, W_T)$ into the drifted one used in the closed-form barrier prices.

## J. PDE routes

**J1.** State the method-of-images identity and the down-and-out call it produces. What is $\alpha$? > Key: for the BS operator $LV=V_t+rSV_S+\frac{1}{2}\sigma^2S^2V_{SS}-rV$, if $LV=0$ then the image $U(S,t)=S^{2\alpha}V(B^2/S,t)$ also solves $LU=0$, with $\alpha=\frac{1}{2}(1-2r/\sigma^2)$. This is the BS analogue of reflecting a heat-equation solution across a wall. Subtracting the image with the right weight cancels the value on the barrier $S=B$, giving the down-and-out call $C_{\text{down-out}}=C_{\text{van}}(S,K) - (S/B)^{2\alpha}\, C_{\text{van}}(B^2/S,K)$: the first term is the vanilla, the second removes exactly the paths that would have crossed B, evaluated at the mirror spot $B^2/S$.

**J2.** *(Derive.)* Sine-series PDE route: what change of variables reduces the localized BS PDE to the heat equation, and why does the sine basis appear? > Key: $\tau=\frac{1}{2}\sigma^2(T-t)$, $\kappa=2r/\sigma^2$, $V=e^{\alpha x+\beta\tau}U$ with $\alpha=-\frac{1}{2}(\kappa-1)$, $\beta=-\alpha^2-\kappa$, $z=x-a \Rightarrow U_\tau=U_{zz}$ on $[0,b-a]$, $U(0,\tau)=U(b-a,\tau)=0$. Homogeneous Dirichlet BCs ⇒ sine series. Propagate modes by $e^{-(n\pi/(b-a))^2\tau}$.

## K. COS for discrete barriers (the main event)

**K1.** *(High priority.)* Why does a discretely-monitored barrier become a BACKWARD RECURSION? Where does the Markov property enter? > Key: survival = $\prod_m 1\{x_m<h\}$; Markov ⇒ joint density factorises into one-step transitions ⇒ the M-dim integral nests ⇒ backward recursion: $c(x,t_{m-1})=e^{-r\Delta t}\int v(y,t_m)f(y|x)dy$, then knock out: $v=R$ for $x \ge h$, $=c$ for $x<h$.

**K2.** Write the COS step for the continuation value. How is it the European COS formula? > Key: $\hat{c}(x,t_{m-1})=e^{-r\Delta t}\sum'_k F_k(x)V_k(t_m)$, $F_k(x)=\mathrm{Re}[\varphi(k\pi/(b-a);x)e^{-ik\pi a/(b-a)}]$, $V_k$=cosine coeffs of $v(y,t_m)$. It IS Euro COS with $v(\cdot,t_m)$ playing the role of the payoff — except v's coefficients come from the previous step.

**K3.** *(Core.)* Why must $V_k$ be propagated by backward induction, and how does the knock-out split $V_k$ at h? > Key: v is piecewise (R above h, $\hat{c}$ below) ⇒ $\hat{V}_k(t_m)=\hat{C}_k(a,h,t_m) + e^{-r(T-t_{m-1})}(2R/(b-a))\Psi_k(h,b)$. $\hat{C}_k$ inserts the COS continuation value ⇒ $\hat{C}_k=e^{-r\Delta t}\mathrm{Re}[\sum'_j \varphi_{Levy}(j\pi/(b-a))V_j(t_{m+1})M_{k,j}]$, with $M_{k,j}$ an analytic coupling integral.

**K4.** *(High priority — contrast with Lecture 6.)* Why does the discrete-barrier COS use the FFT when European COS does not? Give the matrix structure and the complexity. > Key: $M_{k,j}$ splits into a Hankel part (depends on j+k) and Toeplitz part (depends on j-k). Toeplitz/Hankel × vector = convolution = 3 FFTs via circulant embedding ⇒ O(N log N) per step. Euro COS is a single O(N) sum needing no coefficient propagation; the barrier needs all $\{V_k\}$ at every step. Total: $O((M-1)N \log_2 N)$.

**K5.** State the truncation range for the barrier problem. What extra constraint beyond Lecture 6? > Key: $[a,b]=(c_1+x_0) \pm L\sqrt{c_2+\sqrt{c_4}}$, $x_0=\ln(S_0/K)$, cumulants of the one-step increment. Extra: the barrier $h=\ln(H/K)$ must sit well INSIDE [a,b] or the split at h is meaningless; and range is fixed once but reused for all M steps.

**K6.** How is Heston handled? Continuous monitoring? > Key: Heston has a 2-D state (log-stock, variance), so the one-step continuation integral is two-dimensional: either 1-D COS in the log-stock with numerical integration over the variance, or a full 2-D COS recovering the joint density. The characteristic function is Heston's (Lecture 7) and the Feller condition $2\kappa\bar{v} \ge \eta^2$ keeps it well-behaved. Continuous monitoring is the $M\to\infty$ ($\Delta t \to 0$) limit; in practice the discrete price converges to it and a

Richardson-type extrapolation in $\Delta t$ — using the known $O(\sqrt{\Delta t})$ discrete-vs-continuous barrier bias — recovers the continuous price.